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An improved PCA-based multi-channel audio coding scheme

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Abstract

In recent years, multi-channel audio coding has become an important research area with the rapid development of mobile communication. This paper proposes a multi-channel audio coding scheme based on adaptive channel grouping and an improved principal component analysis algorithm. The channel grouping algorithm divides a multi-channel audio signal into multiple channel groups according to the correlation coefficient between channels. The improved principal component analysis averages the data to obtain the covariance matrix and it can concentrate the multi-channel audio signal components on as few channels as possible to achieve a reduction in signal dimensions. The experimental results show that the improved scheme achieves a higher reconstructed audio quality compared with the original scheme.

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Keywords: multi-channel audio coding, adaptive channel grouping, Principal Component Analysis (PCA);

1. Introduction

Multi-channel audio coding is currently widely used in entertainment areas such as music, movies, TV broadcasts and video games. Meanwhile, with the development of mobile communication driven by the 3rd Generation Partnership Project (3GPP), various wireless applications have greatly enriched our daily work and life. Some of these applications (e.g., virtual/augmented reality, wireless home entertainment) need multi-channel digital audio and hence multi-channel digital audio coding technology for mobile communication is becoming an important research area.

Actually, some multi-channel audio coding technologies have become industry standards, e.g., AC-3¹ from Dolby Laboratories, MPEG Audio Layer III (MP3) and Advanced Audio Coding (AAC)² from the Moving Picture

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Experts Group (MPEG) organization³, DTS of Digital Cinema Systems⁴, ATRAC of Sony Corporation and WMA of Microsoft Corporation, etc..

In China, the Standard Working Group for Digital Audio and Video Codecs⁵ proposed a multi-channel coding scheme based on adaptive subspace mapping (called AVS2_PCA). This scheme uses Principal Component Analysis (PCA)⁶ to map multi-channel audio signals to optimized subspace, and then code the new channel signal and mapping model⁷. It takes into account both the statistical characteristics and psychoacoustic characteristics of multi-channel audio signals, and it can effectively improve the efficiency of multi-channel audio coding.

This paper proposes a channel grouping algorithm and uses an improved PCA algorithm to implement an optimized multi-channel audio coding scheme. The rest of the paper is organized as follows: Section 2 describes the original PCA-based multi-channel audio coding scheme. Section 3 introduces the application of the channel grouping algorithm and improved PCA in an improved scheme. Experimental results and conclusions are given in Sections 4 and 5 respectively.

2. The PCA-based multi-channel audio coding scheme

The PCA-based multi-channel audio coding scheme is the AVS2_PCA scheme, which uses the PCA method for subspace mapping. The encoding framework is shown in Fig. 1.

Firstly, the multi-channel audio signal is pre-processed, including channel separation and resampling, and the resampled signal is output to the signal type module in frame. Then the Modified Discrete Cosine Transform (MDCT) maps the input digital audio signal to the MDCT domain after the classification. PCA analysis is performed on the multi-channel MDCT spectral coefficients of each PCA group by using a PCA module to obtain PCA principal component MDCT spectral coefficients and PCA coefficient matrix. Among them, the selected PCA principal component is mono-coded to obtain coded data of the PCA principal component; the PCA coefficient matrix is quantized and encoded. Finally, the encoded data of the PCA principal component, the encoded data of the PCA coefficients and the side information are multiplexed to form an audio encoding code stream.

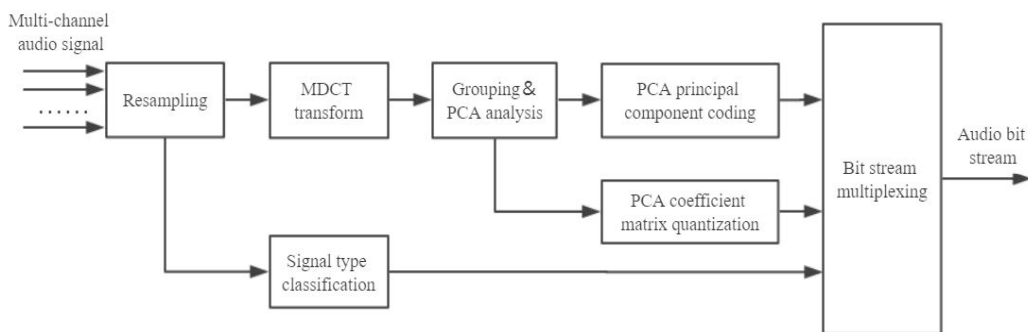


Fig. 1. AVS2_PCA audio encoding framework⁷

3. The improved multi-channel audio coding scheme

Firstly, the framework of the improved multi-channel audio coding scheme is described. Secondly, aiming at solving the problems caused by data standardization in PCA, an improved PCA algorithm is given and adopted in the audio coding scheme. And finally an adaptive channel grouping scheme is described.

3.1. The framework of the improved coding scheme

Compared with the original coding scheme, there are two differences in the improved multi-channel audio coding scheme. One is an adaptive grouping algorithm and the other is the improved PCA. Meanwhile, Enhance Voice

Services (EVS) codec is adopted to encode the PCA principal component in both the original scheme and the improved one. The encoding framework is shown in Fig. 2.

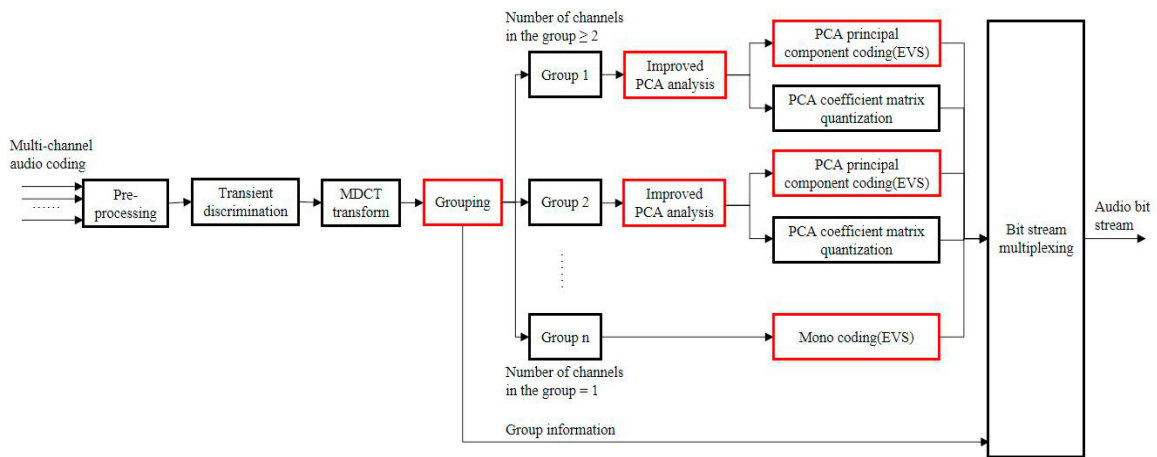


Fig. 2. The improved multi-channel audio encoding scheme

3.2. The improved PCA and its experimental results

As to the PCA method, Shuangfeng Ye studied some deficiencies in its practical application and then proposed corresponding improvement methods⁸. The main conclusions are as follows: (1) While data standardization eliminates the impact of dimensionality, it also smoothes out the difference information in the degree of data variation. (2) PCA is just a "linear" dimensionality reduction technique and can only deal with linear problems.

Aiming at solving the problems caused by data standardization, we use data averaging for optimization and the following improvements are proposed. The raw data is averaged $z_{ij} = x_{ij} / \bar{x}_j$. Among them, the original data is

$(x_{ij})_{n \times p}$, the average value of each column is $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$, $j = 1, 2, \dots, p$. Let z_{ij} be the data after the data x_{ij} averaged. The remaining steps are the same as the traditional PCA method. The detailed approach is shown in Table 1.

Table 1. Improved PCA Algorithm

i. Combine raw data into a matrix A .
ii. Divide the raw data by its column mean to obtain matrix A' .
iii. Calculate the covariance matrix after averaging.
iv. Calculate eigenvalues $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ and eigenvectors $a_1, a_2, a_3, \dots, a_n$ of W .
v. Construct a matrix B after sorting eigenvectors.
vi. Select the top k columns of B according to the cumulative contribution rate to get B' .
vii. Calculate $C = A' * B'$.

The eigenvalues, contribution rates and cumulative contribution rates of the correlation coefficient matrix R of the original data after standardization (before improvement) and the covariance matrix V of the original data after averaging (after improvement) are obtained through experiments. The experimental results are shown in Fig. 3.

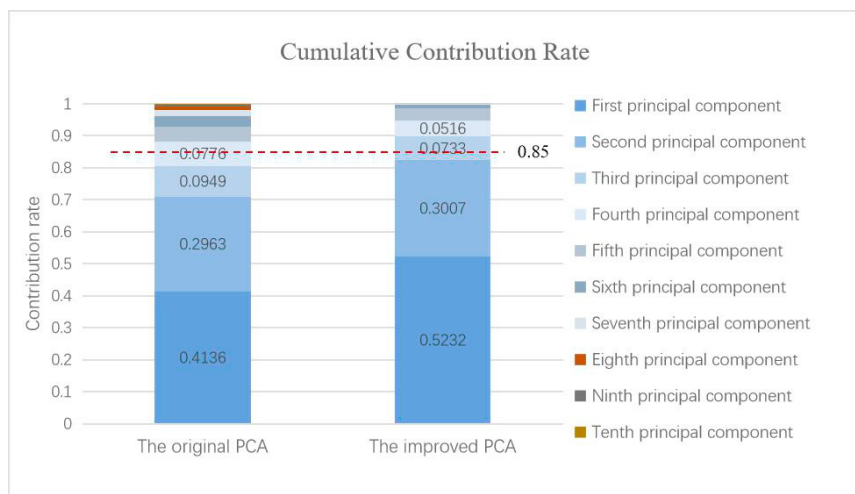


Fig. 3. Comparison of the cumulative contribution rate between the original PCA and the improved PCA

From Fig. 3, it can be seen that the contribution rate of the first principal component after improvement is 52.32% which is 11% higher than before. According to the principle that the cumulative contribution rate is not less than 85%, 3 principal components should be selected after improvement while 4 principal components are selected before. Therefore, the averaging method can extract more information with fewer principal components.

3.3. Channel grouping algorithm

A new multi-channel audio grouping algorithm is proposed based on cluster analysis. The main idea is to group channels according to the correlation of sound signals between channels. Specifically, if the correlation coefficient between the two channels p and q is bigger than the threshold (the threshold is set to 0.5), then the channels p and q are grouped into the same channel grouping, on the contrary make p and q be different groupings. The side information of each subband includes the number of groups and the sequence number of the channels contained in each group. The detailed steps are shown in Table 2.

Table 2. Channel Grouping Algorithm

i. Combine raw data into a matrix S .
ii. Calculate the correlation coefficient d between any two channels in S and save it as a two-dimensional array.
iii. Find the largest correlation coefficient d_{ab} and the corresponding two-channel m_a, m_b .
iv. If $d_{ab} \geq 0.5$, create a group X , add m_a, m_b to X and remove them from S ; otherwise, channel grouping cannot be performed.
v. Calculate the correlation coefficient between any channel in S and each channel in X . If both d are ≥ 0.5 , put the channel into X , otherwise X is an independent grouping and contains only m_a and m_b .
vi. Repeat steps iv-vi in the newly generated S until each channel in S is a group.
vii. Record the side information of each frequency domain subband.

3.4. Bit allocation

Since this scheme adopts an adaptive grouping method, the coding rate is different for each frame in different speech materials, in other words, its bit rate is variable. The bit allocation of the improved coding scheme is shown in Table 3.

Table 3. Bit Allocation of the Improved Coding Scheme

Parameters		Number of bits
Principal component		960
Coefficient matrix	Two-dimensional	5
	Three-dimensional	18
	four-dimensional	27
Side information	Band	6
	Group	3-21
Total		3207-3364

4. Experimental results and analysis

4.1. Experiment design

This experiment uses the MUSHRA (Multi-Stimulus Test with Hidden Reference and Anchor) subjective test method⁹ in order to evaluate the reconstructed audio quality of 5.1-channel audio signals at 48kHz sampling rate under different conditions.

The test sequence consists of 7 kinds audio signals, including mechanical sound, music, game sound (English), movie sound (English), movie sound (Chinese) and instrument sound.

The test audio materials are the original signal, the anchor signal, the reconstructed audio signal of the original scheme, the reconstructed audio signal of the improved scheme and the mono-codec signal by EVS Codec. The bit rate of the both the original scheme and the improved scheme is 167.4 kbps and the mono coding signal by EVS Codec is 192 kbps.

There are 16 subjects, including 8 males and 8 females with normal hearing.

4.2. Test results

The test results of MUSHRA obtained are shown in Fig. 4, and the 95% confidence intervals of each codec are [85.2, 93.0], [54.8, 67.6], [75.0, 83.8], [76.5, 87.1], [73.8, 80.8].

It can be seen that the MUSHRA score of the improved multi-channel audio codec is 81.8, which is higher than the original coding scheme at the same bit rate. It is because that the improved PCA comprehensively extracts valid information from the original data. What's more, PCA method can make full use of the statistical characteristics of the multi-channel audio signal and reduce statistical redundant information between channels. Therefore, compared with individual channel audio coding by EVS, using the PCA technique in the multi-channel audio coding scheme achieves better coding results with a lower bit rate.

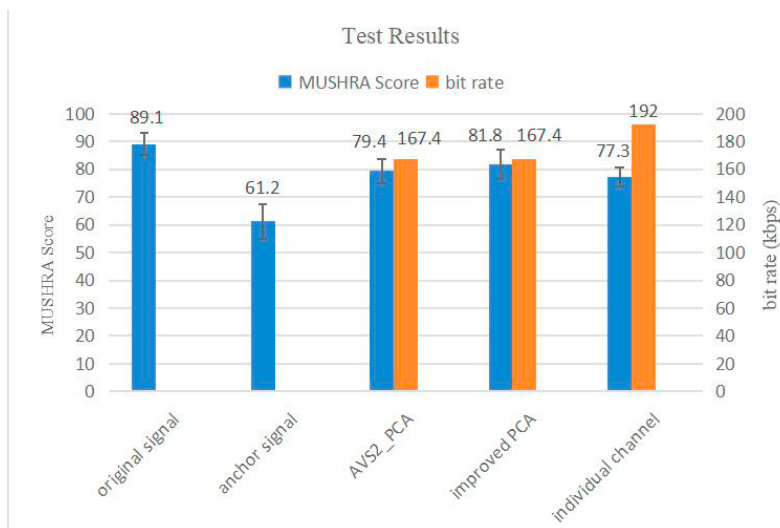


Fig. 4. Test results

5. Conclusion

This paper proposes an improved multi-channel audio coding scheme, which is based on adaptive channel grouping and an improved principal components analysis algorithm. The experimental results show that the quality of the reconstructed audio signal with the improved multi-channel audio coding scheme is higher than the original scheme. Further work includes the optimal design of the original matrix before PCA and the quantization of some coding parameters.

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